

Jyotsna Ramanan

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Tech professional with *8+ years* of experience in Computer Science, consistently delivering impactful analytics solutions across *10+* projects empowering stakeholders to make informed decisions. Proven ability to develop and deliver *5+* high-quality web development projects, and a demonstrated aptitude for research with *2+* publications

EXPERIENCE

• Learning Analyst

September 2019-Present

University of Nebraska System

Lincoln, NE

- Integrate, mine, and model *large-scale* LMS, SIS *datasets* showcasing actionable insights on student performance and behavior, therefore **empowering stakeholders** to make informed decisions about students
- Primary analyst on all phases of creation of in-house learning analytics LTI product called **Course Insights**
- Received highest possible performance rating ‘**Exceeds Expectations**’ during the evaluation term
February 2022-February 2023
- Present demos and share insights on current work with peers at **BTAA** and **Unizin** institutions, **fostering collaboration across academia**
- Engage across NU System to communicate identified Learning analytics opportunities to faculty, administration, and data stewards
- Co-presented a case-study finding at EDUCAUSE Annual Conference 2020, **Evaluating Student Behavior and Outcomes in Online Testing**
- Regularly utilize strong **analytical** and **problem-solving skills**, **statistical analysis**, and **programming knowledge** by leveraging technologies like **Tableau**, **R** (*Libraries: TidyR, Dplyr, SNA*), **Python** (*Libraries: NumPy, Pandas, Matplotlib, Seaborn*) and **SQL** (*MS SQL Server, PostgreSQL, Google BigQuery*), **Statistics** (*Descriptive, Probability, z-scores, hypothesis testing, ANOVA, Pearson correlation, linear regression*), and **GitLab**

• Software Engineer

April 2017-September 2019

Resource Pro LLC

Lincoln, NE

- Worked on **5+ projects** to create transformative streamlined software solutions that automate commercial insurance processes such as new business, renewals, and account management, saving time for account managers and insurance service representatives by **70% to 90%**
- Worked individually on R&D of proof-of-concepts using **machine learning** in Azure ML and Microsoft Cognitive Services for creating tools such as **product recommender** and **insurance chat bot**
- **Collaborated extensively** with peers in overseas development team (Bangalore, India), bridging the gap effectively between US-based stakeholders and them, and vice-versa
- Mentored and provided technical guidance to a Junior UI Developer (from team in Bangalore, India)
- Implemented features in company’s Salesforce environment using **APEX**, **SOQL** and **VisualForce** on Classic and Lightning environments
- Worked closely with and *reported directly* to the VP of Technology
- Profound use of technologies like **ASP.NET MVC 5** with **C#**, **HTML5**, **Bootstrap with CSS3**, **JavaScript**, **jQuery**, and various other JavaScript libraries, and **MS SQL Server**, and **BitBucket** for version control

• Teaching Assistant

January 2016-December 2016

School of Computing, University of Nebraska-Lincoln

Lincoln, NE

- Held office hours at least twice a week to meet and guide students with any questions they have regarding the assignments/projects
- Assisted faculty instructors with **creation of course materials** like coursework content, assignments, quizzes, exams, & projects, and with **grading** subsequent student responses
- Courses TA’d:
 - Spring 2016: CSCE 913 - Advanced Topics in Database Systems, CSCE 430/830 - Computer Architecture
 - Fall 2016: CSCE 413/813 - Database Systems, CSCE 471/870 - Computer Graphics (WebGL Programming)

• Software Developer Intern

May 2015-December 2015

DataBankIMX, LLC

Lincoln, NE

- Rapidly acquired proficiency in web development in **ASP.NET MVC 5 Framework** with **C#** Entity Framework, and **MS SQL Server 2008**, **HTML5** and **CSS Bootstrap 2** for full-stack web development
- Successfully created responsive and engaging full stack web portals to manage electronic documents for notable local clients

• Data Analyst-Student Worker

September 2014-May 2015

Dept. of Child, Youth, and Family Studies, University of Nebraska-Lincoln

Lincoln, NE

- Performed **data management** and **analyses** on assessment data received via website **GoNapSacc**
- Used technologies like **MS-Excel** for data cleaning & pre-processing, and **IBM-SPSS** for statistical analyses as a part of a research project.

EDUCATION

- **Post-Baccalaureate (Part-time)**

University of Nebraska-Lincoln

Spring 2021, Spring 2024

Course: **Statistical Methods** (EDPS 859)

Final Grade and GPA: **A|4.0**

Course: **Intro to Data Mining** (CSCE 874)

Final Grade and GPA: **A|4.0**

Capstone Project: Connecting Genomic Data to Phenotypic Expression in Maize A project featuring biological data set of a panel of maize inbred plants' **genotype profile** and **phenotype expressions**, to derive meaningful associations, including the **determination of maize yield** levels(high, medium, low) in relation to these observed phenotype traits. I actively contributed to perform **discretization** on the voluminous data set, and subsequently conducted **association rule mining** to generate item sets of key features such as *cob length*, *ear mass*, *weight of kernels* and their relevant *yield* category mentioned earlier. **View on GitLab**

- **MS in Computer Science**

University of Nebraska-Lincoln

Fall 2014-Fall 2016

GPA: 3.6

Thesis: Testing the Independence Hypothesis of Accepted Mutations for Pairs of Amino Acids in Protein Sequences

A computer-based study to test if the pairwise mutations of adjacent amino acid pairs in protein sequences are independent of each other under the assumption that single mutations are usually independent of each other, according to evolutionary studies **Verified by WES**

RESEARCH PUBLICATIONS

- **Conference Paper**

J.Ramanan, P.Z.Revesz, *Mutations of Adjacent Amino Acid Pairs are not Always Independent*, INASE Conference, Crete Island, Greece, October 17-19, 2015

- **Journal Paper**

J.Ramanan, P.Z.Revesz, *Testing the independence hypothesis of accepted mutations of adjacent amino acids in protein sequences*, International Journal of Biology and Biomedical Engineering, 11, 170-179, 2017

CERTIFICATES

- **Lean Six Sigma-Yellow Belt**

February 11, 2019

- **Machine Learning by Stanford University on Coursera**

April 16, 2019